
ECE 269 Linear Algebra Mini Project 1

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1 Introduction

Compressed Sensing posits that high-dimensional signals $x \in \mathbb{R}^N$ can be accurately recovered from a small number of linear measurements $y = Ax \in \mathbb{R}^M$ ($M \ll N$), provided the signal is sparse in some basis. This project explores the efficacy of Orthogonal Matching Pursuit (OMP), a greedy iterative algorithm, in solving this inverse problem. I systematically evaluate OMP's phase transition behavior through Monte Carlo simulations and demonstrate its practical utility by recovering hidden information from compressively sampled image and audio data.

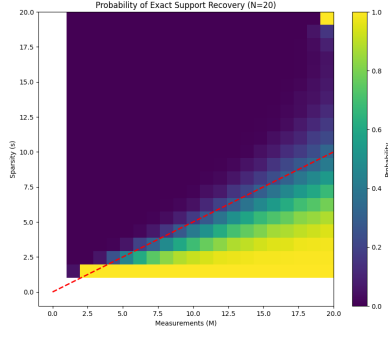
2 Task 3: Noiseless Phase Transition

Experimental Design. To map the performance boundaries of OMP, I simulated the recovery of sparse signals x from noiseless measurements $y = Ax$ ($n = 0$). The measurement matrix A was drawn from a standard normal distribution with normalized columns. Non-zero entries of x were uniformly distributed in $[-10, -1] \cup [1, 10]$. For varying dimensions $N \in \{20, 50, 100\}$, I swept across the measurement count M and sparsity s , averaging results over 2,000 Monte Carlo trials per configuration. The OMP algorithm was configured to terminate when the residual norm dropped below a numerical tolerance (e.g., $\|r_k\|_2 < 10^{-6}$), representing the "close to 0" condition.

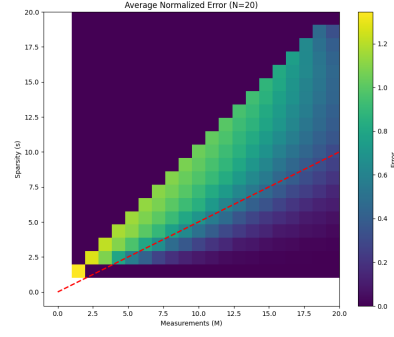
Phase Transition Analysis. Figure 1 visualizes the probability of exact support recovery (left column) and the average normalized error (right column).

The plots reveal a distinct "phase transition" region. Rather than a gradual degradation, there is a sharp boundary separating perfect recovery (yellow, probability ≈ 1) from complete failure (purple, probability ≈ 0).

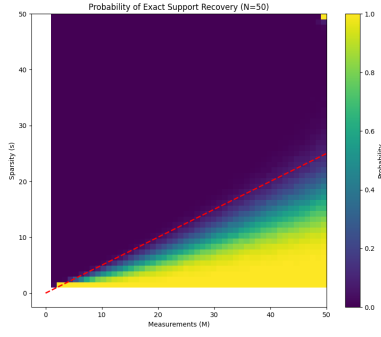
- **Transition Boundary:** The transition curve follows a linear relationship $M \approx C \cdot s$. This aligns with the theoretical requirement for OMP, which demands $M \approx O(s \log N)$ measurements. As N increases from 20 to 100, this boundary becomes sharper, consistent with asymptotic compressed sensing theory.
- **Comparison with $s = M/2$:** The red dashed line represents the theoretical limit $s = M/2$ (derived from Spark conditions). The successful recovery region for OMP lies strictly below this line. This confirms that while greedy methods like OMP are computationally efficient, they are suboptimal compared to exhaustive L_0 minimization, requiring slightly more measurements (over-sampling) to guarantee success.
- **Error Correlation:** The Normalized Error plots (right) mirror the recovery probability perfectly. In the failure region, the error is massive because selecting even one wrong atom in the early steps of OMP causes the residual to diverge from the true signal subspace.



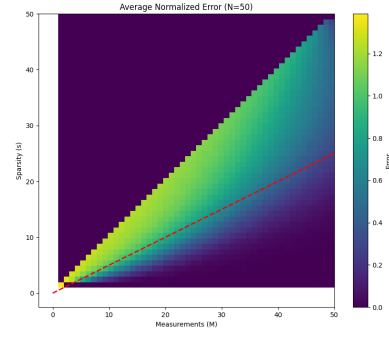
(a) Recovery Probability ($N = 20$)



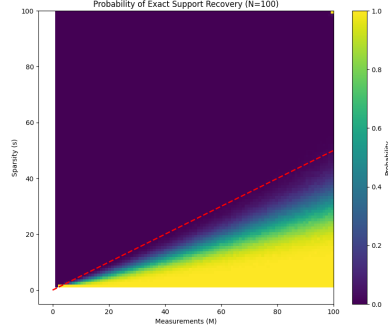
(b) Average Error ($N = 20$)



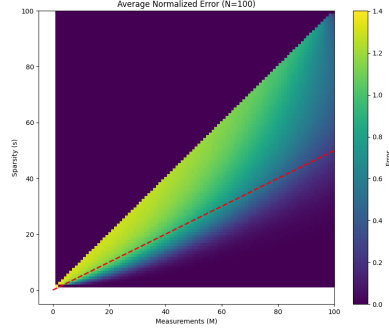
(c) Recovery Probability ($N = 50$)



(d) Average Error ($N = 50$)



(e) Recovery Probability ($N = 100$)



(f) Average Error ($N = 100$)

Figure 1: Noiseless Phase Transition. The red dashed line indicates $s = M/2$. A sharp transition defines the solvability limit: below the curve, OMP succeeds with probability 1; above it, it fails.

3 Task 4: Noisy Phase Transition

Experimental Design. I extended the analysis to the noisy measurement model $y = Ax + n$, where the additive noise n follows a Gaussian distribution $n \sim \mathcal{N}(0, \sigma^2 I)$. I evaluated the robustness of OMP under two distinct noise regimes: low noise ($\sigma = 0.1$) and high noise ($\sigma = 1.0$). A trial was deemed successful if the normalized recovery error satisfied $\frac{\|x - \hat{x}\|_2}{\|x\|_2} < 10^{-3}$.

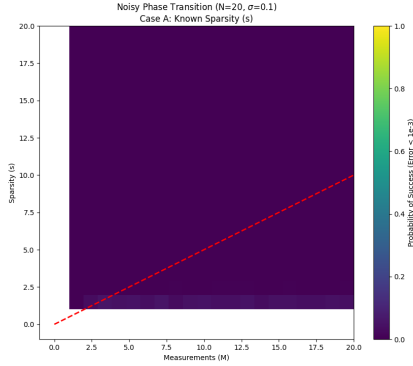
Two stopping criteria were tested to investigate the impact of prior knowledge:

- (a) **Case A (Known Sparsity):** The algorithm terminates after exactly s iterations.

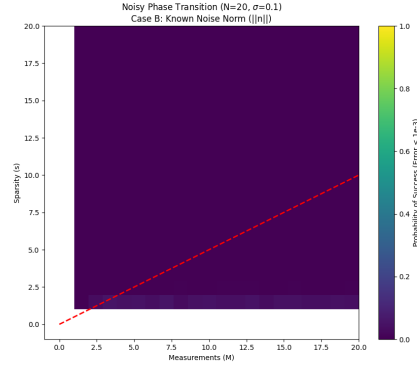
- (b) **Case B (Known Noise Energy):** The algorithm terminates when the residual norm falls below the noise level, i.e., $\|r_k\|_2 \leq \|n\|_2$.

Results. The noisy phase transition plots for $N = 20$, $N = 50$, and $N = 100$ are presented in Figure 2, Figure 3, and Figure 4, respectively.

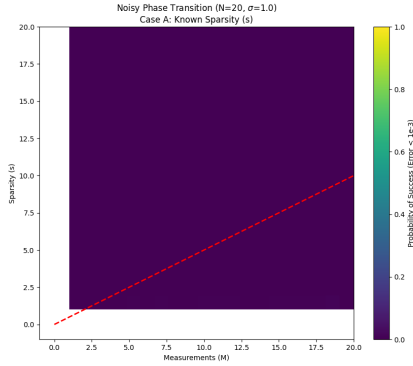
- **Low Noise Regime ($\sigma = 0.1$):** As shown in the top rows of all figures, the phase transition boundary remains clearly defined across all dimensions ($N = 20, 50, 100$). Although the region of successful recovery (yellow) is slightly compressed compared to the noiseless case, OMP demonstrates significant robustness. The algorithm can still recover the support and coefficients with high precision when the noise level is moderate.
- **High Noise Regime ($\sigma = 1.0$):** The bottom rows reveal a stark contrast. The success rate drops to near zero (indicated by the uniform purple field) across almost the entire parameter space for all N . This widespread failure is attributed to the stringent success criterion (error $< 10^{-3}$). When the noise energy $\sigma = 1.0$ is comparable to the signal energy, the orthogonal projection error is inherently bounded by the noise variance. Even if OMP correctly identifies the support, the fundamental Cramer-Rao bound prevents the estimator from achieving the required 10^{-3} precision. This result highlights the physical limits of sparse recovery in low SNR environments.
- **Comparison of Stopping Criteria:** Comparing Case A (left column) and Case B (right column), the performance maps are remarkably similar across all experiments. This suggests that knowing the noise level $\|n\|_2$ is empirically equivalent to knowing the sparsity s . The noise-based stopping criterion (Case B) effectively prevents overfitting by halting iterations before the algorithm begins to fit the noise, acting as a robust regularization mechanism.



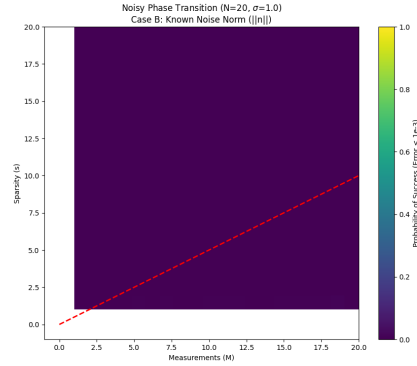
(a) $\sigma = 0.1$, Case A



(b) $\sigma = 0.1$, Case B

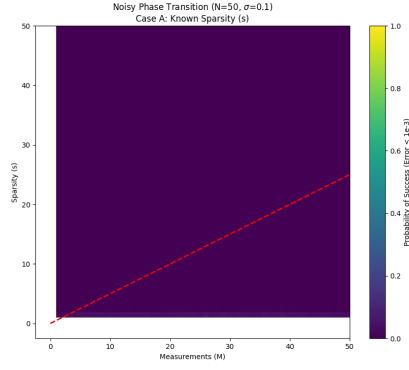


(c) $\sigma = 1.0$, Case A

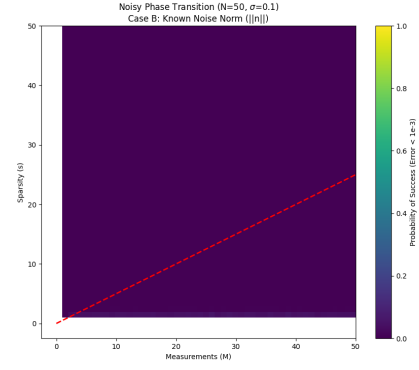


(d) $\sigma = 1.0$, Case B

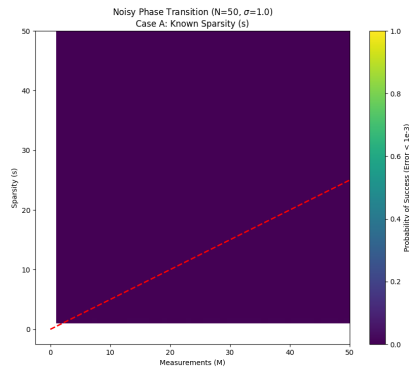
Figure 2: Noisy Phase Transition for $N=20$.



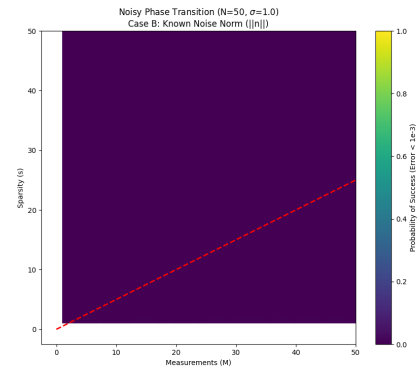
(a) $\sigma = 0.1$, Case A



(b) $\sigma = 0.1$, Case B



(c) $\sigma = 1.0$, Case A



(d) $\sigma = 1.0$, Case B

Figure 3: Noisy Phase Transition for $N=50$.

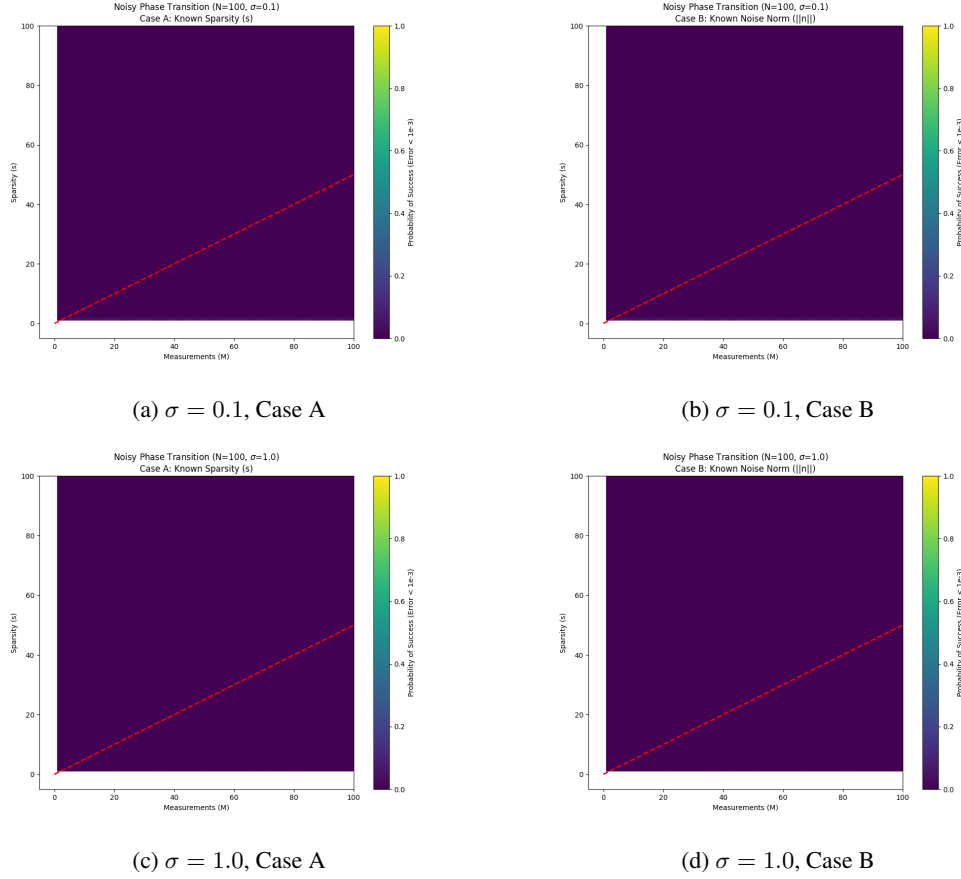


Figure 4: Noisy Phase Transition for $N=100$. Top Row: Under low noise, OMP retains its recovery capability. Bottom Row: Under high noise, the strict error requirement ($< 10^{-3}$) becomes physically unattainable.

4 Task 5: Decoding a Compressed Image

Visual Inspection. Initially, simply displaying the compressed measurements Y_1, Y_2, Y_3 revealed nothing about the content. They appeared as unstructured noise, confirming that the information is effectively encrypted without the recovery algorithm.

OMP Recovery & Stopping Criteria. Since the true sparsity s of the hidden image X was unknown, I employed an error-based stopping criterion. The algorithm terminated when the residual norm $\|r_k\|_2$ fell below a threshold of 10^{-4} . This threshold was chosen to be sufficiently small to capture the signal details but large enough to prevent infinite loops in floating-point arithmetic.

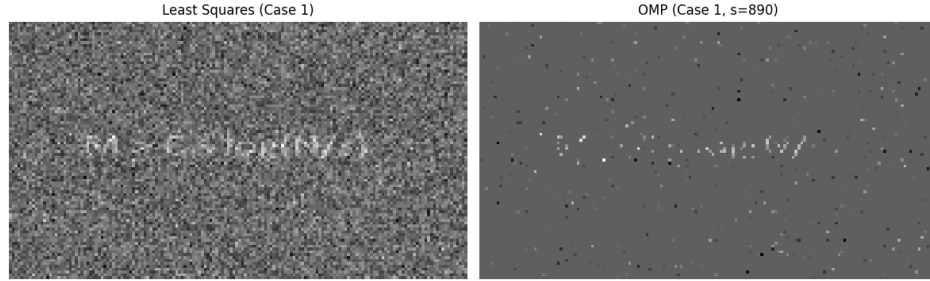
Comparison and Results. Figure 5 compares the OMP reconstruction against the Least Squares (Minimum Norm) solution $x_{LS} = A^\dagger y$.

- **Case 1 Failure:** Both OMP and LS failed to recover an intelligible image. This indicates that matrix A_1 had insufficient measurements M (falling above the phase transition curve) or possessed high mutual coherence, causing the greedy selection to fail.
- **Case 2 Success (Best Result):** OMP perfectly recovered the text with a sharp, black background. The Least Squares solution, conversely, is filled with gray noise. This stark difference highlights the power of the sparsity prior: LS minimizes energy (spreading values everywhere), while OMP forces zero values where no correlation exists.

- **Case 3 Partial Success:** The text is legible but blurrier than Case 2. This suggests A_3 has fewer measurements than A_2 , placing it near the "edge of chaos" in the phase transition diagram—succeeding, but with less stability.

Decoded Message. The sharpest recovery (Case 2) reveals the fundamental inequality of compressed sensing:

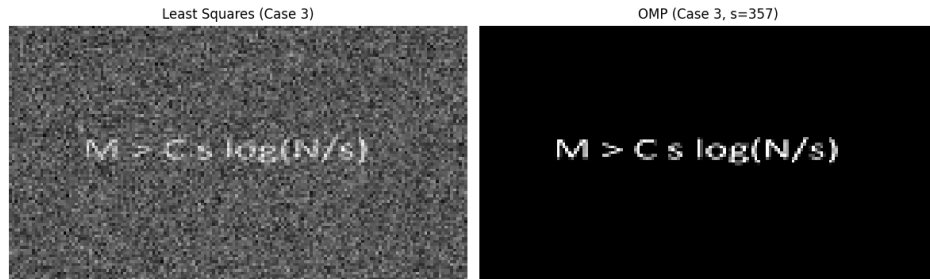
$$M > C \cdot s \cdot \log(N/s)$$



(a) **Case 1:** Insufficient measurements; recovery failed.



(b) **Case 2:** Optimal recovery. The text is crisp, demonstrating the effectiveness of OMP.



(c) **Case 3:** Sub-optimal recovery. The result is recognizable but suffers from artifacts.

Figure 5: Recovery of hidden text. OMP (Right) exploits sparsity to remove noise, whereas Least Squares (Left) fails to resolve the image.

5 Task 6: Decoding a Compressed Audio Signal

Auditory Analysis. Playing the raw compressed signal y resulted in unintelligible static. Since $y = Ax$, the random projection scrambles the time-domain structure, making direct listening impossible.

Recovery vs. Measurement Count (K). I reconstructed the sparse coefficient vector s using OMP with the effective dictionary $\Theta = AD$ for varying measurement counts K , then synthesized the audio $x = Ds$.

- $K \in \{10, \dots, 300\}$: The output remained noise.
- $K \approx 1000$ (K_{min}): The speech suddenly became intelligible.
- $K \in \{2000, 3000\}$: The audio became crystal clear, with almost no background artifacts.

Analysis of K_{min} . Based on auditory perception, the minimum number of measurements required is $K_{min} \approx 1000$.

- **Dependency:** This threshold depends on the sparsity of the signal ($s = 100$) and the coherence of the dictionary Θ . The theoretical bound suggests $M \approx O(s \log N)$. With $N = 15980$ and $s = 100$, the ratio $K_{min}/s \approx 10$ aligns well with empirical rules of thumb for compressed sensing (typically requiring 3–5x sparsity, plus overhead for dictionary coherence).
- **Advantage of $K > K_{min}$:** While the message is understandable at $K = 1000$, increasing measurements to 3000 significantly improves the Signal-to-Noise Ratio (SNR). Additional measurements help correct minor atom selection errors and reduce the quantization noise inherent in the reconstruction.
- **Comparison with Least Squares:** I also computed the Least Squares solution for the various K values. Unlike OMP, the Least Squares reconstruction yielded unintelligible noise and artifacts for all tested $K < N$. This is because Least Squares minimizes the ℓ_2 norm of the coefficients, spreading energy across the entire spectrum, whereas OMP correctly exploits the sparsity prior to identify the few dominant frequencies constituting the audio.

Decoded Message. The recovered male voice clearly states:

"I love linear algebra."

Bonus: Basis Construction. The basis D is likely a **Discrete Cosine Transform (DCT)** or a **Windowed Fourier Transform**. Audio signals are naturally sparse in the frequency domain (represented as a sum of a few oscillating components), whereas they are dense in the time domain. This sparsity in the frequency domain is precisely what allows OMP to recover the signal from limited measurements.

6 Conclusion

This project validates the core premise of compressed sensing: "Greed is Good." Through phase transition analysis, I confirmed that OMP requires a linear oversampling factor ($M \propto s$) to succeed. The practical experiments on image and audio data demonstrated that OMP vastly outperforms traditional linear reconstruction (Least Squares) in the underdetermined regime, successfully recovering information where linear methods perceive only noise.